

Satellite Launch System: Digital Twin Mission

A project of a launch system for delivering a small satellite to low Earth orbit (LEO), covering rocket design, flight profile, and separation sequence, while creating a digital twin that simulates and reproduces all these elements and enables mission trajectory analysis, including rocket mass optimization while still achieving orbit.

Challenge Description

The objective is to create a digital twin of a launch system for delivering a small satellite to low Earth orbit (LEO).

The project should include:

- launch vehicle design,
- rocket flight model,
- mission profile,
- separation sequence,
- flight simulation,
- analysis of achieving the target orbit.

Participants are expected to design a simplified system that enables pre-flight mission simulations and analysis of how rocket parameters affect mission success.

Project Scope

The project may be implemented in any technology and form, for example:

- desktop application,
- web application,
- simulation environment,
- computational script/program,
- integration with OpenRocket or a similar tool.

Permitted tools include:

- OpenRocket,
- Scilab,
- Python,

- MATLAB/Simulink,
- Unity,
- and other software as needed.

Minimum Requirements

The project should include:

1. Rocket Model

At a minimum, it should account for:

- mass,
- engine,
- basic aerodynamic parameters,
- rocket structure.

2. Flight Simulation

It should include:

- flight trajectory,
- altitude,
- speed,
- fuel consumption,
- flight stability.

3. Mission Sequence

For example:

- launch,
- stage separation,
- payload separation,
- orbit achievement.

4. Digital Twin Element

The system should enable:

- modification of rocket parameters,

- re-running simulations,
- analysis of how changes affect mission progression.

Additional advanced features that can improve project evaluation

- 3D visualization,
- telemetry dashboard,
- AI/ML for parameter optimization,
- interactive user interface,
- other creative ideas from participants that improve solution quality.

Constraints

The project does not need to reproduce full orbital physics at a professional or industrial level. The key factors are concept, creativity, and a working prototype.

Simplified mathematical and simulation models are acceptable.

Submission Format

The team should prepare:

- a short presentation of the solution,
- a description of how the system works,
- a simulation demonstration.

Optional:

- technical documentation,
- demo recording.

Evaluation Criteria

Evaluation will consider:

- innovation of the solution,
- correctness of simulation behavior,
- quality of implementation,
- technical level of the project,
- creativity,

- quality of project presentation.

Useful Links

<https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/guide-to-rockets/#fluid-dynamics>

<https://science.nasa.gov/learn/basics-of-space-flight/>

<https://openrocket.info/> + <https://openrocket.readthedocs.io/en/latest/>

<https://www.scilab.org/>

<https://ocw.mit.edu/courses/16-346-astrodynamics-fall-2008/>

Files on Disk

[https://drive.google.com/drive/folders/1XmJd7AMY OCD2ZDj0mEOAFkvq0DeOhGJK?](https://drive.google.com/drive/folders/1XmJd7AMY OCD2ZDj0mEOAFkvq0DeOhGJK?usp=drive_link)

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